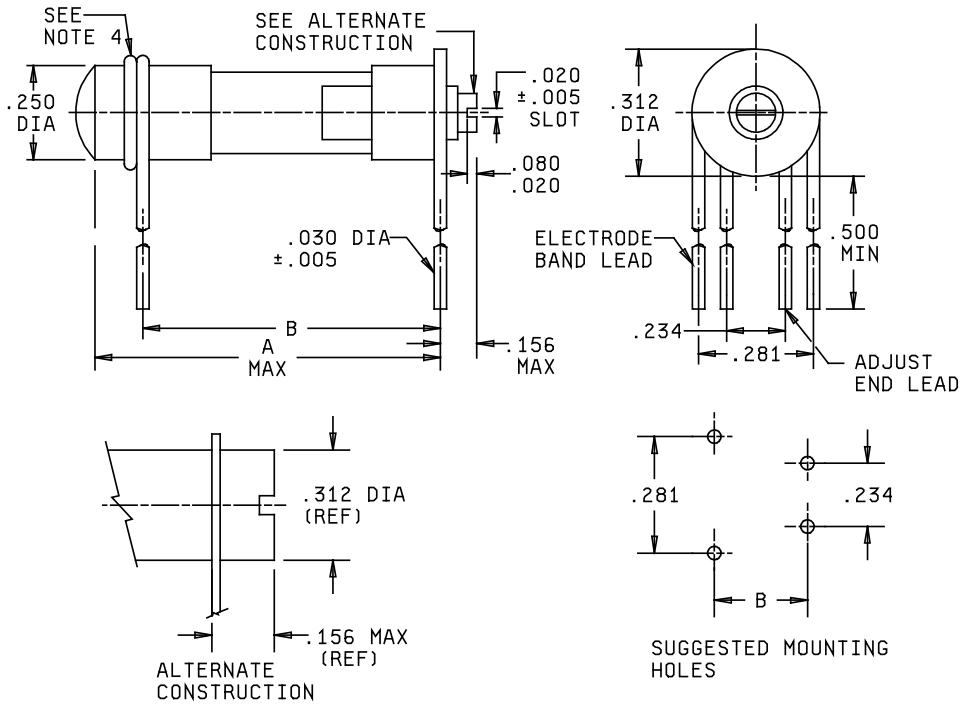
INCH-POUND
MIL-PRF-14409/8F
23 February 2024
SUPERSEDING
MIL-PRF-14409/8E
6 December 2018

PERFORMANCE SPECIFICATION SHEET

CAPACITOR, VARIABLE (PISTON TYPE, TUBULAR TRIMMER), STYLES PC42 AND PC52

This specification sheet is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-14409.



Inches	mm
.005	0.13
.016	0.41
.020	0.51
.030	0.76
.080	2.03
.156	3.96
.234	5.94
.250	6.35
.281	7.14
.312	7.92
.500	12.70

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are in millimeters and are given for general information only.
3. Unless otherwise specified, tolerance is ± .016 inch.
4. There shall be a minimum of one and one-half turns of wire around the body. These turns of wire shall be completely soldered to the body.
5. This style capacitor is designed for printed-circuit mounting.
6. For dimensions A and B, see table I.

FIGURE 1. Dimensions and configuration.

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REQUIREMENTS:

Dimensions: See figure 1 and table I.

DC voltage rating: See table I.

Construction: Sealed glass or quartz, see figure 1 and table I.

Insulation resistance (IR): MIL-STD-202-302, test condition B (500 volts ±10 percent); not less than 1,000,000 megohms, except characteristics J and H shall be not less than 100,000 megohms at +125°C.

Capacitance range: See table I.

Quality factor (Q): See table I.

Thermal shock: MIL-STD-202-107, test condition A, except the step 3 temperature shall be +125°C ± 5°C for characteristics J and H, and +150°C ± 5°C for characteristic Q.

Immersion: MIL-STD-202-104, test condition A.

Seal test: Applicable.

TABLE I. Capacitor characteristics.

Part or identifying number (PIN)	DC rated voltage	Capacitance range		Dimensions 1/		Q	Dielectric material	Wire size
		Min	Max	B ± .031 (0.79)	A (max)			AWG
	Volts	pF	pF					
PC42Q1R8	750	0.6	1.8	.500 (12.70)	.656 (16.66)	1,500	Quartz	24
PC42Q5R5	1,250	0.6	5.5	.703 (17.86)	.922 (23.42)	1,500	Quartz	22
PC42Q9R5	1,250	0.6	9.5	1.016 (25.81)	1.359 (34.52)	1,500	Quartz	22
PC42Q160	1,250	0.8	16.0	1.469 (37.31)	1.953 (49.61)	1,500	Quartz	22
PC52J5R5	750	0.8	5.5	.500 (12.70)	.656 (16.66)	1,000	Glass	24
PC52J110	1,250	0.8	11.0	.703 (17.86)	.906 (23.01)	900	Glass	22
PC52H160	1,250	0.8	16.0	.844 (21.44)	1.109 (28.17)	800	Glass	22
PC52H230	1,250	0.8	23.0	1.016 (25.81)	1.359 (34.52)	700	Glass	22
PC52H380	1,250	0.8	38.0	1.469 (37.31)	1.953 (49.61)	500	Glass	22

1/ Metric equivalents are given for general information only. Millimeters are in parentheses.

TABLE II. Substitutability data.

Superseded type designation in MIL-C-14409/8	Substitutable type designation in MIL-PRF-14409/8A through /8F
PC42J4R5	PC52J5R5
PC42J8R5	PC52J110
PC42H120	PC52H160
PC42H180	PC52H230
PC42H300	PC52H380

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Changes from previous issue: The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

Referenced documents: In addition to MIL-PRF-14409, this document also references:

MIL-STD-202-104
MIL-STD-202-107
MIL-STD-202-302

Custodians:
Army - CR
Navy - EC
Air Force - 85
DLA - CC

Preparing activity:
DLA - CC

(Project 5910-2024-019)

Review activities:
Navy - AS, MC, OS

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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